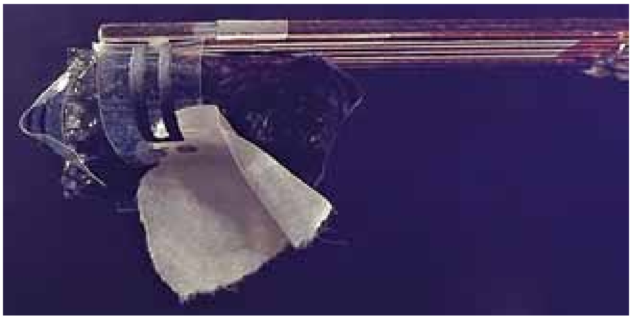
Soft robotic hand for future surgical applications

In the realm of surgical robotics, soft robots hold promise for enhancing precision and flexibility. Traditional rigid robots face limitations in water-rich environments like the human body, prompting TU/e researchers to pioneer a new design. They have developed a soft robotic gripper from organic materials, featuring liquid crystals and graphene. This innovation aims to revolutionise surgical robots, offering safer and more adaptable tools for procedures. Soft robotics, leveraging materials like fluids and elastics, seeks to overcome the constraints of rigid metals in robotic performance. The integration of these advancements could significantly impact surgical precision and expand capabilities in minimally invasive operations.



Soft robotic hand for future surgical applications (Credit: www.tue.nl/en)

Renewable energy transition may not reduce net energy

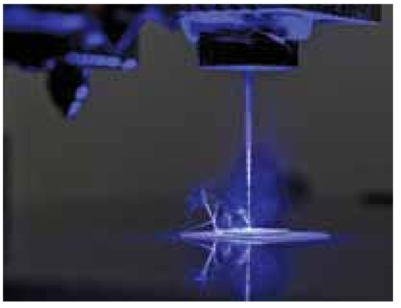
Researchers at the University of Leeds explored concerns over the net energy output of renewable energy versus fossil fuels. They aimed to determine if transitioning to sustainable energy technologies reduces net energy production. The findings suggest that the shift to renewables may not necessarily decrease net energy availability. Unlike previous studies that focused on initial extraction for fossil fuels and final delivery for renewables, recent research, including a 2019 Nature Energy article, indicates a narrowing gap when assessing energy returns at the same final stage.

A transceiver boosts 5G coverage in blocked areas

Tokyo Tech scientists have developed a groundbreaking 256-element transceiver array for non-line-of-sight (NLoS) 5G communication. This wirelessly powered innovation excels in power transmission and conversion efficiency, promising expanded 5G network coverage in obstructed areas. By overcoming link blockages caused by obstacles like buildings, this advancement enhances communication flexibility and coverage scope. It holds potential to democratise access to high-speed, low-latency communication crucial for next-gen wireless networks. Operating at millimetre-wave frequencies (24-100GHz), it tackles challenges of signal-to-noise ratios (SNR) and obstruction in current 5G setups. The transceiver's design integrates gallium arsenide diodes, baluns, DPDT switches, and digital ICs, marking a significant stride towards robust NLoS 5G deployment.

No assembly needed with 3D printing in manufacturing

Researchers at the University of Missouri have developed a pioneering method using a single machine to create complex devices from multiple materials such as plastics, metals, and semiconductors. Known as the freeform multi-material assembly process, this innovation utilises advanced 3D printing and laser techniques to produce multi-material sensors, circuit boards, and textiles embedded with electronics. It integrates sensors into structures, enabling objects to detect environmental conditions like temperature and pressure. This breakthrough promises applications in oceanic research sensors and wearable health monitors. The process streamlines prototyping complexities, fostering advancements in wearable sensors, customisable robots, and medical devices.



University of Missouri researchers have built a machine that combines elements of traditional 3D printing with laser technology to develop multi-material, multi-functional products (Credit: https://showme.missouri.edu)

Wafer-scale 2D material integration with van der Waals contacts

Researchers from the University of Science and Technology of China (USTC) have pioneered an all-stacking method for achieving robust 2D van der Waals (vdW) contacts. This technique directly stacks metal electrodes onto 2D materials without traditional metal deposition, safeguarding material integrity. Published in Nature Communications, their study underscores enhanced device operability and scalability. Devices employing this approach exhibit precise metal-semiconductor interfaces and clear vdW gaps, with no metal atom doping on the 2D material side. Transistors fabricated this way demonstrate a 95% reduction in off-state current and improved on-off ratios, crucial for low-power circuits.